

PROJECT DOCUMENTATION

Smart Leave Management System

Abstract

The Smart Leave Management System is a ServiceNow-based application developed to automate the employee leave request process within an organization. The system allows employees to submit leave requests through a custom application. The application routes requests for manager approval, automatically calculates leave duration, validates user inputs, and provides dashboard reporting for request monitoring.

The project reduces manual effort, improves transparency, and provides a centralized platform for tracking employee leave requests.

Introduction

Employee leave management is an essential process in every organization. Traditional methods of managing leave requests often involve emails, spreadsheets, or paper-based forms, which may lead to delays, calculation errors, and lack of visibility into request status.

To overcome these challenges, the Smart Leave Management System was developed using ServiceNow. The system automates leave request submission, approval workflows, leave duration calculation, validation management, and reporting.

Problem Statement

Many organizations still use manual processes to manage employee leave requests.

Challenges include:

- Manual leave request handling
- Approval delays
- Lack of request tracking
- Errors in leave duration calculation
- Increased administrative workload
- Communication gaps

These challenges reduce efficiency and increase processing time.

Objectives

The objectives of the project are:

Primary Objectives

- Automate employee leave requests.
- Reduce paperwork and manual intervention.
- Improve approval efficiency.
- Enable leave request tracking.
- Automate leave duration calculation.
- Provide dashboard reporting.

Secondary Objectives

- Improve user experience.
- Ensure process transparency.
- Maintain leave request history.
- Reduce calculation errors.

Development Environment

Platform Used

ServiceNow Developer Instance

Development Modules

- ServiceNow App Studio
- Flow Designer
- Business Rules
- Client Scripts
- Custom Tables
- Dashboards and Reports
- Form Designer

Application Development

Step 1: Creation of Application

A custom application named:

Smart Leave Management System

was created using ServiceNow App Studio.

Purpose

The application serves as a centralized platform for managing employee leave requests.

Application Components

- Custom Table
- Leave Request Form
- Approval Flow
- Business Rule
- Client Script
- Dashboard

4. Custom Table Development

A custom table was created to store employee leave request information.

Table Name

Leave Request

Fields Created

Field Name | Description

Employee Name | Employee requesting leave

Leave Type | Type of leave requested

Start Date | Leave start date

End Date | Leave end date

Total Days | Leave duration

Reason | Purpose of leave

Approval Status | Approval state

Purpose

The table stores all leave request records and acts as the primary data repository for the application.

5. Form Design Development

A user-friendly form was designed for employee leave request submission.

Form Fields

Employee Information

- Employee Name

Leave Information

- Leave Type
- Start Date
- End Date
- Total Days
- Reason

Validation

Mandatory fields were configured to ensure complete leave request submission.

Benefits

- Improved usability
- Reduced data entry errors
- Standardized leave request collection

6. Leave Request Form Development

A custom leave request form was created to allow employees to submit leave requests.

Form Name

Leave Request Form

Variables Created

Variable | Type

Employee Name | Single Line Text

Leave Type | Choice

Start Date | Date

End Date | Date

Total Days | Integer

Reason | Multi-Line Text

Approval Status | Choice

Functionality

Employees can submit leave requests using a structured form.

7. Workflow Development Using Flow Designer

A workflow was created using Flow Designer to automate the leave request lifecycle.

Flow Name

Leave Approval Flow

Trigger

Leave Request Created

The workflow starts automatically whenever a new leave request record is created.

Approval Action

An approval step was configured using:

Ask For Approval

Purpose

The request is routed to the reporting manager for review and approval.

Approval Rule

Anyone Approves

Conditional Logic

An IF condition was configured.

Condition

Approval State = Approved

Purpose

The workflow proceeds only when approval is granted.

Status Management

The request status is automatically updated based on the manager's decision.

Approved → Status = Approved

Rejected → Status = Rejected

8. Business Rule Development

A Business Rule was configured to automatically calculate the total number of leave days.

Business Rule Name

Calculate Total Leave Days

Purpose

Automatically calculates leave duration based on the selected Start Date and End Date.

Example

Start Date : 23-Jun-2026

End Date : 25-Jun-2026

Total Days : 3

Benefits

- Eliminates manual calculations
- Improves accuracy
- Reduces processing time

9. Client Script Development

A Client Script was configured to validate leave dates before submission.

Client Script Name

Leave Date Validation

Validation Rule

End Date cannot be earlier than Start Date.

Error Message

"End Date cannot be earlier than Start Date"

Purpose

Prevents invalid leave requests and improves data quality.

10. Dashboard Development

A dashboard was created to monitor leave requests and approval status.

Dashboard Components

- Total Requests
- Approved Requests
- Rejected Requests
- Pending Requests

Purpose

Provides real-time visibility into leave request activities and approval trends.

Results

The project successfully automated the employee leave request process.

Achievements

- Automated leave request submission
- Automated approval workflow
- Automatic leave duration calculation
- Client-side validation
- Dashboard-based monitoring
- Centralized leave management

Benefits

- Reduced processing time
- Increased transparency
- Improved efficiency
- Better tracking and reporting
- Improved data accuracy

Conclusion

The Smart Leave Management System successfully automates the complete employee leave request lifecycle using ServiceNow. The application provides a structured workflow for leave request submission, approval management, automatic leave calculation, validation, and dashboard reporting. By reducing manual effort and improving process visibility, the system enhances organizational efficiency and user satisfaction.